Name:_____ ID:_____

CSE 2046/2246 Analysis of Algorithms

Spring 2022 Final

Duration: 105 minutes

Q1	Q2	Q3	Q4	Q5	Q6	Q7	SUM
/18	/22	/10	/20	/16	/20	/14	/100

Q-1. (18 pts) Mr. Varyemez is trying to put together a holiday gift knapsack for Donald Duck. He has n items to choose from, each with infinitely many copies (aka. knapsack with replacement). Item i has weight w_i, and value v_i. Mr. Varyemez wants to pick some items (possibility with duplicates) so their total weight is exactly W, while minimizing the total value of the items picked. (Note that w_1 is always equal to 1, and W, w_i and v_i are integer values. So there is always a solution.)

(a – 8 pts) Our goal is to design a dynamic programming algorithm. Write a recurrence relation for F(w) which is the minimum total value needed to make a weight of w. Give the base conditions as well.

$F(w) = \min_{w_j \le w} \{ F(w - w_j) + v_j \}$

$F(1) = v_1$

$F(0) = 0$

(b – 8 pts) Briefly describe a dynamic programming algorithm using the recurrence relation in part (a) and apply your algorithm for the following instance:

$W = 6$ and there are three items:
$w_1 = 1, v_1 = 3$
$w_2 = 3, v_2 = 4$
$w_3 = 5, v_3 = 6$

Starting from F(2), we fill F(i) according to the recurrence relation. We increment i by one, and stop after finding F(6).

$F(1) = 3$

$F(2) = \min\{F(2-1)+3\} = 6$

$F(3) = \min\{F(3-1)+3, F(3-3)+4\} = 4$

$F(4) = \min\{F(4-1)+3, F(4-3)+4\} = 7$

$F(5) = \min\{F(5-1)+3, F(5-3)+4, F(5-5)+6\} = 6$

$F(6) = \min\{F(6-1)+3, F(6-3)+4, F(6-5)+6\} = 8$

We can find the items by going backwards

item 1, item 1

(c – 2 pts) Describe time and space complexity of your algorithm

Time: $O(Wn)$

Space: $\Theta(W)$.

Name:_____ ID:_____

Q-2. (22 pts) Consider the same problem in Q-1.
(a – 7 pts) Describe a greedy algorithm for this problem.

We sort the items according to $\frac{v_i}{w_i}$.
Then choose the least valuable one as long as there is space. If there is no space choose the least valuable one that fit the knapsack.

(b – 5 pts) Apply your algorithm for the same instance.
$W = 6;\ \ w_1 = 1, v_1 = 3\ ;\ \ w_2 = 3, v_2 = 4\ ;\ \ w_3 = 5, v_3 = 6$

$\frac{v_1}{w_1} = 3 \quad \frac{v_2}{w_2} = \frac{4}{3} \quad \frac{v_3}{w_3} = \frac{6}{5}$

Choose item 3. Remaining capacity is 1.
Cannot choose item 2.
Choose item 1. Remaining capacity is zero.

(c – 3 pts) What is the accuracy ratio of your algorithm for the above instance?

Total value is now $6+3=9$

Accuracy ratio $= \frac{9}{8}$ //

(d – 7 pts) Show that the performance ratio of your algorithm is unbounded above. (Hint: you may try to arbitrarily increase one of the values in the above instance.)

If we increase v_1 to 3000, accuracy ratio will be $\frac{3006}{8}$.
Because we choose it without any option.

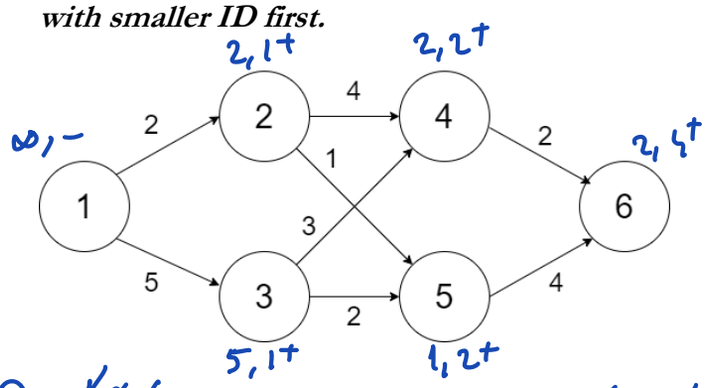
Q-3. (10 pts) Explain how Warshall's algorithm can be used to determine whether a given digraph is a dag (directed acyclic graph). Is it a good algorithm for this problem? Explain.

We apply Warshall's algorithm & find the transitive closure.
Then if there is a 1 in the diagonal, then it means that this is not a dag. 7pts
Complexity of this algorithm is $\Theta(n^3)$. 3pts.
There exist algorithms with $O(n^2)$ time complexity, so it is not a good algorithm.

Name:_____ ID:_____

Q-4. (20 pts) Max Flow Problem

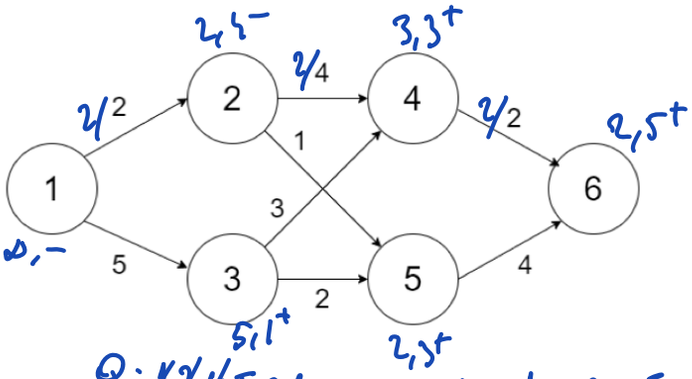
(a – 15 pts) Apply the shortest augmenting path algorithm to find the maximum flow (source is 1 and sink is 6). Show all the labels and BFS queues. ***While applying BFS, visit the neighbor with smaller ID first.***



Q: 1 2 4 5 6

aug path: 1→2→4→6

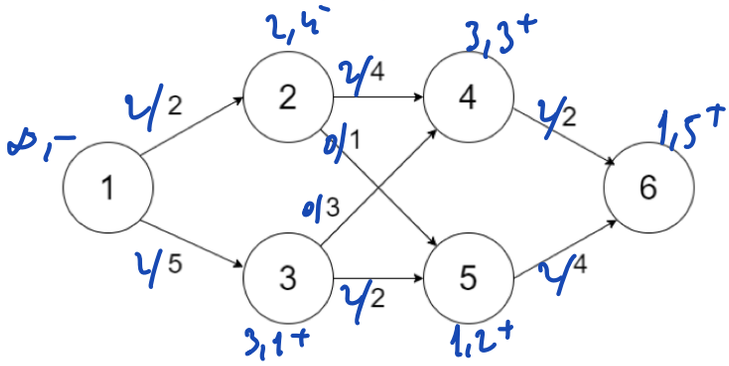
flow: 2



Q: 1 3 4 5 2 6

aug. path: 1→3→5→6

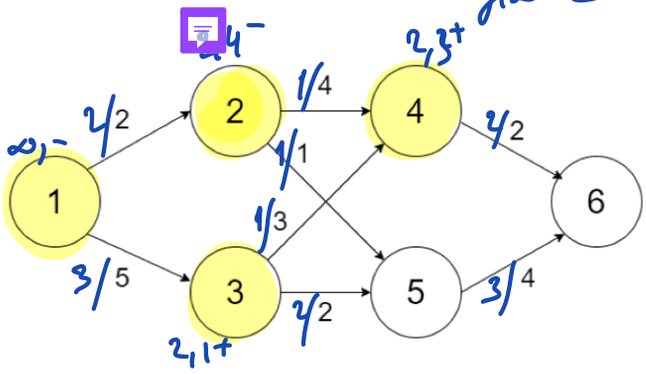
flow: 2



Q: 1 3 4 2 5 6

aug path: 1→3→4←2→5→6

flow: 1



Q: 1 3 4

total flow: 5

(b – 5 pts) According to the labels in the final step, describe the minimum cut found by the algorithm.

X: Labelled: 1, 2, 3, 4

$\bar{X}$: Unlabelled: 5, 6

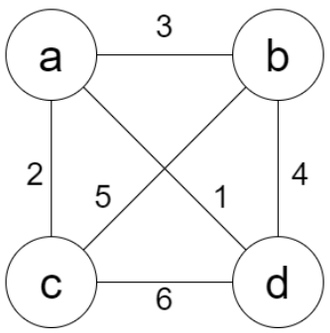
min cut = {(4,6), (2,5), (3,5)}

value of min cut = 5

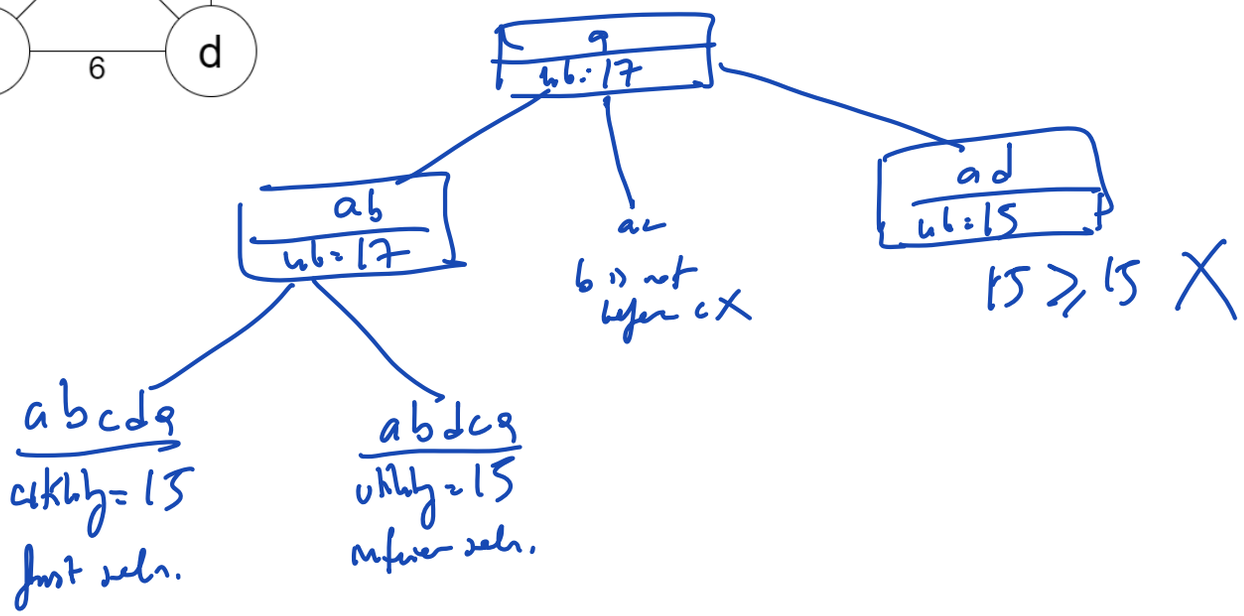
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Q-5. (16 pts) Branch-and-Bound

Consider a weighted undirected graph G, where weigths of edges are the utilities. Now, consider the version of Traveling Salesman Problem called MaxTSP, where the goal is to find a tour with maximum utility (not minimum cost as we did in the class.). Describe a bounding function and apply the branch-and-bound algorithm to find the tour with maximum utility in the following graph: (Note: Your answer will not be graded if you solve for standard TSP.)



$ub: \left\lfloor \frac{\overbrace{3+2}^{a} + \overbrace{5+4}^{b} + \overbrace{6+5}^{c} + \overbrace{6+4}^{d}}{2} \right\rfloor = \left\lfloor \frac{35}{2} \right\rfloor = 17$



$ab: ub = \left\lfloor \frac{3+2+3+5+6+5+6+4}{2} \right\rfloor = \left\lfloor \frac{34}{2} \right\rfloor = 17$

$ad: ub = \left\lfloor \frac{4+3+5+4+6+5+6+4}{2} \right\rfloor = \left\lfloor \frac{31}{2} \right\rfloor = 15$

Name:_____ ID:_____

Q-6. (20 pts) Lower Bound, P, NP.

(a – 6 pts) What is the trivial lower bound class of finding most distant numbers among n real numbers $x_1, x_2, ..., x_n$? Is this lower bound tight? Why?

Trivial LB: $\Omega(n)$ because we have to process all the inputs

Yes it is tight, because we need to find max & min, both of which can be done in linear time.

(b – 5 pts) Given that decision version of standard TSP problem is NP-Complete, prove that decision version of maxTSP described in Q-5 is also NP-Complete (Note: Decision version can be defined as "deciding whether there exist a tour with a utility higher than K").

max TSP is in NP, because it can be verifiable in poly time.

We need to reduce TSP to max TSP, which can be done by only all the utilities by -1.

(c – 9 pts) Suppose that Problem X can be reduced to Problem Y, and Problem Y can be reduced to the Sorting problem. Problem Y can also be reduced to Problem Z.
Which of the following are definitely true? Describe. True answers without any description will not get any point.

i. **Problem X is tractable.**

True, because it can be solved in $O(n \lg n)$ by reducing to sorting problem

ii. **Problem Z is tractable.**

We don't know. Problem Z can be a difficult problem.

iii. **Problem Z is at least as hard as Problem X.**

Yes. Problem Z is at least as hard as Y, which is at least as hard as X.

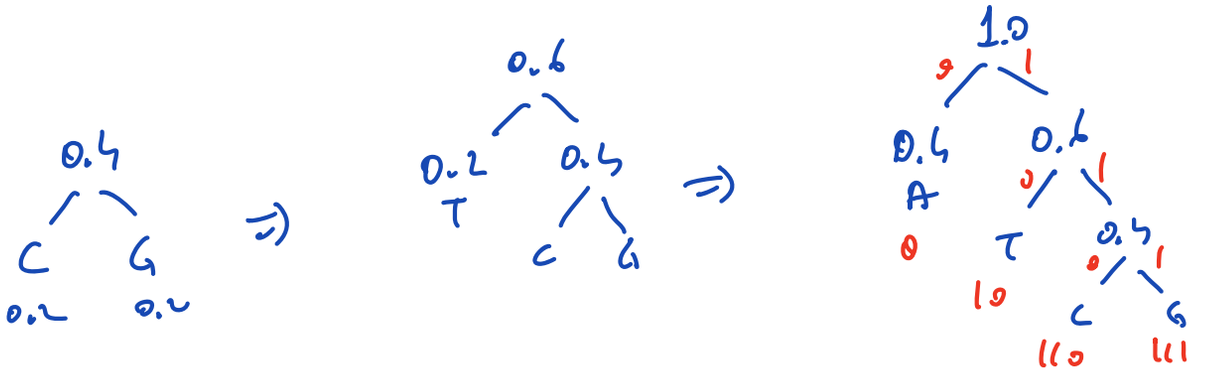
Name:_____ ID:_____

Q-7. (14 pts) Suppose that a file consists of 4 different characters A, C, G and T. Frequency of A is known to be 40%. What is the minimum possible compression ratio of Huffman coding algorithm compared to a fixed size encoding? Assign appropriate frequency values to C, G, and T such that the compression ratio is minimized. Then solve the coding problem and find the codewords and the compression ratio.

A : 0,4

We need to maximize the frequency of least occuring characters in order to minimize the compression ratio.

A: 0.4 C: 0,2 G: 0,2 T: 0,2 is such a case.



Avg codeword length: $0,4 \cdot 1 + 0,2 \cdot 2 + 0,2 \cdot 3 + 0,2 \cdot 3$

$= 0,4 + 1,6 = 2$

Compression: $\frac{2-2}{2} = 0\%$.